

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 2(a + 3b) =$$

$$(2) \quad 5(4x + 7) =$$

$$(3) \quad 2(4x - 3) =$$

$$(4) \quad 4(2y + 1) =$$

$$(5) \quad 4(3a + 1x) =$$

$$(6) \quad 5(a + 2y) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 4(b - 2) =$$

$$(2) \quad -2(x + 1a) =$$

$$(3) \quad 3(2y + 1) =$$

$$(4) \quad 3(a + 6) =$$

$$(5) \quad 5(2x + 3y) =$$

$$(6) \quad 4(y - 4) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 2(b + 2x) =$$

$$(2) \quad -2(x + 1y) =$$

$$(3) \quad -4(3a - 4b) =$$

$$(4) \quad -5(x - 1a) =$$

$$(5) \quad 5(a - 6) =$$

$$(6) \quad 5(y + 4) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 5(2x + 2) =$$

$$(2) \quad 2(3y - 4b) =$$

$$(3) \quad 4(2a + 2) =$$

$$(4) \quad 2(3a - 1) =$$

$$(5) \quad 5(2x + 5) =$$

$$(6) \quad 2(x - 1b) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 4(x - 3a) =$$

$$(2) \quad 2(3b + 8) =$$

$$(3) \quad 4(4b + 4) =$$

$$(4) \quad 4(2a + 4) =$$

$$(5) \quad 3(3a + 2) =$$

$$(6) \quad 4(y - 3a) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 4x + 4(4x + 1y) =$$

$$(2) \quad 4(x + 2) + 3(x - 5) =$$

$$(3) \quad 4(y + 3) + 3(y - 5) =$$

$$(4) \quad 2(y + 4) + 4(y + 1) =$$

$$(5) \quad 4(a + 4) + 4(a + 4) =$$

$$(6) \quad 3(x + 1) + 4(x - 1) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 4(b + 2) + 4(b + 2) =$$

$$(2) \quad 3a + 4(4a + 1y) =$$

$$(3) \quad 4y + 3(-1y + 2x) =$$

$$(4) \quad 4(x + 4) + 4(x - 4) =$$

$$(5) \quad 4(x + 4) + 2(x - 2) =$$

$$(6) \quad 2a + 2(3a + 3y) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 4a + 4(-1a + 4b) =$$

$$(2) \quad 2y + 3(-1y + 2b) =$$

$$(3) \quad 2y + 3(-3y + 2x) =$$

$$(4) \quad 3(b + 1) + 4(b + 3) =$$

$$(5) \quad 4(a + 1) + 2(a - 1) =$$

$$(6) \quad 4(a + 2) + 3(a + 5) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 3(y + 2) + 2(y + 3) =$$

$$(2) \quad 5(a + 4) + 4(a + 3) =$$

$$(3) \quad 4(a + 4) + 3(a + 1) =$$

$$(4) \quad 2(x + 3) + 2(x + 2) =$$

$$(5) \quad 4y + 3(-1y + 1a) =$$

$$(6) \quad 3(b + 4) + 2(b - 5) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (a + 3)(a - 3) =$$

$$(2) \quad (y + 4)^2 =$$

$$(3) \quad (x - 4)^2 =$$

$$(4) \quad (x - 3)^2 =$$

$$(5) \quad (a + 4)(a - 4) =$$

$$(6) \quad (a - 7)^2 =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (a + 3)(a - 3) =$$

$$(2) \quad (a + 6)^2 =$$

$$(3) \quad (x + 8)^2 =$$

$$(4) \quad (a + 7)^2 =$$

$$(5) \quad (a - 6)^2 =$$

$$(6) \quad (y - 2)^2 =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 5a + 3(2a + 1b) =$$

$$(2) \quad 3y + 3(3y + 1a) =$$

$$(3) \quad 5(x + 1) + 2(x - 4) =$$

$$(4) \quad 4y + 3(3y + 4x) =$$

$$(5) \quad 5(x + 1) + 4(x + 1) =$$

$$(6) \quad 3(a + 4) + 4(a + 5) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (y + 6)^2 =$$

$$(2) \quad (a + 4)^2 =$$

$$(3) \quad (y + 7)(y - 7) =$$

$$(4) \quad (y - 2)^2 =$$

$$(5) \quad (x - 4)^2 =$$

$$(6) \quad (a - 6)^2 =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (a - 6)^2 =$$

$$(2) \quad (x + 3)^2 =$$

$$(3) \quad (x + 6)^2 =$$

$$(4) \quad (y + 8)^2 =$$

$$(5) \quad (x + 8)(x - 8) =$$

$$(6) \quad (a - 5)^2 =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (y + 6)(y + 6) =$$

$$(2) \quad 4b(3b - 4) =$$

$$(3) \quad (x - 4)(x + 2) =$$

$$(4) \quad 4b(1b + 1) =$$

$$(5) \quad 3a(2a + 6) =$$

$$(6) \quad 3y(1y - 3x) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (y + 3)(y - 3) =$$

$$(2) \quad (x + 5)^2 =$$

$$(3) \quad (y - 2)^2 =$$

$$(4) \quad (y + 4)(y - 4) =$$

$$(5) \quad (x + 3)(x - 3) =$$

$$(6) \quad (x + 2)(x - 2) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 3b(3b + 6x) =$$

$$(2) \quad 5b(3b - 2y) =$$

$$(3) \quad 4x(1x - 3) =$$

$$(4) \quad 2x(3x + 3b) =$$

$$(5) \quad (x + 3)(x - 5) =$$

$$(6) \quad 4x(2x - 3a) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 4y(3y + 6) =$$

$$(2) \quad 4b(2b + 3y) =$$

$$(3) \quad (y - 3)(y + 1) =$$

$$(4) \quad 4b(3b - 4y) =$$

$$(5) \quad (y - 6)(y - 5) =$$

$$(6) \quad 2y(3y + 4x) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad (y - 1)(y - 2) =$$

$$(2) \quad (a + 2)(a + 5) =$$

$$(3) \quad 3a(1a - 1) =$$

$$(4) \quad 3a(2a + 6) =$$

$$(5) \quad 5x(3x - 3) =$$

$$(6) \quad (a - 3)(a + 5) =$$

Section 7. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

$$(1) \quad 3y(4y - 4a) =$$

$$(2) \quad 3x(2x + 4) =$$

$$(3) \quad (x + 4)(x + 6) =$$

$$(4) \quad 2y(1y + 3) =$$

$$(5) \quad (a - 2)(a + 5) =$$

$$(6) \quad (y + 6)(y + 3) =$$

Section 7 - Solutions

61a-1: $4x^2 + 14x$

61a-2: $y^2 - 6y + 8$

61a-3: $x^2 - 1$

61a-4: $6a^2 + 27a$

61a-5: $4y^2 + 36y$

61a-6: $8a^2 - 28a$

61b-1: $4x^2 + 14x$

61b-2: $5a^2 - 40a$

61b-3: $12x^2 + 36x$

61b-4: $6a^2 + 27a$

61b-5: $4y^2 + 36y$

61b-6: $8a^2 - 28a$

62a-1: $4x^2 + 14x$

62a-2: $y^2 - 6y + 8$

62a-3: $x^2 - 1$

Section 7 - Solutions (continued)

63b-1: $4x^2 + 14x$

63b-2: $y^2 - 6y + 8$

63b-3: $12x^2 + 36x$

63b-4: $6a^2 + 27a$

63b-5: $y^2 - 8y + 12$

63b-6: $x^2 - 6x + 9$

64a-1: $4x^2 + 14x$

64a-2: $5a^2 - 40a$

64a-3: $x^2 - 1$

64a-4: $y^2 + 2y - 8$

64a-5: $4y^2 + 36y$

64a-6: $8a^2 - 28a$

64b-1: $4x^2 + 14x$

64b-2: $y^2 - 6y + 8$

64b-3: $12x^2 + 36x$

Section 7 - Solutions (continued)

64b-4: $6a^2 + 27a$

64b-5: $y^2 - 8y + 12$

64b-6: $x^2 - 6x + 9$

65a-1: $4x^2 + 14x$

65a-2: $y^2 - 6y + 8$

65a-3: $12x^2 + 36x$

65a-4: $y^2 + 2y - 8$

65a-5: $y^2 - 8y + 12$

65a-6: $8a^2 - 28a$

65b-1: $y^2 + 1y - 2$

65b-2: $y^2 - 6y + 8$

65b-3: $12x^2 + 36x$

65b-4: $6a^2 + 27a$

65b-5: $4y^2 + 36y$

65b-6: $8a^2 - 28a$

Section 7 - Solutions (continued)

62a-4: $6a^2 + 27a$

62a-5: $y^2 - 8y + 12$

62a-6: $8a^2 - 28a$

62b-1: $4x^2 + 14x$

62b-2: $5a^2 - 40a$

62b-3: $x^2 - 1$

62b-4: $6a^2 + 27a$

62b-5: $y^2 - 8y + 12$

62b-6: $x^2 - 6x + 9$

63a-1: $4x^2 + 14x$

63a-2: $5a^2 - 40a$

63a-3: $x^2 - 1$

63a-4: $y^2 + 2y - 8$

63a-5: $y^2 - 8y + 12$

63a-6: $x^2 - 6x + 9$

Section 7 - Solutions (continued)

66a-1: $4x^2 + 14x$

66a-2: $5a^2 - 40a$

66a-3: $12x^2 + 36x$

66a-4: $y^2 + 2y - 8$

66a-5: $y^2 - 8y + 12$

66a-6: $x^2 - 6x + 9$

66b-1: $y^2 + 1y - 2$

66b-2: $5a^2 - 40a$

66b-3: $x^2 - 1$

66b-4: $y^2 + 2y - 8$

66b-5: $4y^2 + 36y$

66b-6: $x^2 - 6x + 9$

67a-1: $4x^2 + 14x$

67a-2: $y^2 - 6y + 8$

67a-3: $12x^2 + 36x$

Section 7 - Solutions (continued)

68b-1: $y^2 + 1y - 2$

68b-2: $y^2 - 6y + 8$

68b-3: $x^2 - 1$

68b-4: $6a^2 + 27a$

68b-5: $y^2 - 8y + 12$

68b-6: $8a^2 - 28a$

69a-1: $y^2 + 1y - 2$

69a-2: $y^2 - 6y + 8$

69a-3: $x^2 - 1$

69a-4: $6a^2 + 27a$

69a-5: $4y^2 + 36y$

69a-6: $x^2 - 6x + 9$

69b-1: $y^2 + 1y - 2$

69b-2: $y^2 - 6y + 8$

69b-3: $12x^2 + 36x$

Section 7 - Solutions (continued)

69b-4: $6a^2 + 27a$

69b-5: $y^2 - 8y + 12$

69b-6: $8a^2 - 28a$

70a-1: $y^2 + 1y - 2$

70a-2: $y^2 - 6y + 8$

70a-3: $12x^2 + 36x$

70a-4: $y^2 + 2y - 8$

70a-5: $y^2 - 8y + 12$

70a-6: $8a^2 - 28a$

70b-1: $4x^2 + 14x$

70b-2: $y^2 - 6y + 8$

70b-3: $12x^2 + 36x$

70b-4: $y^2 + 2y - 8$

70b-5: $4y^2 + 36y$

70b-6: $8a^2 - 28a$

Section 7 - Solutions (continued)

67a-4: $6a^2 + 27a$

67a-5: $4y^2 + 36y$

67a-6: $x^2 - 6x + 9$

67b-1: $4x^2 + 14x$

67b-2: $y^2 - 6y + 8$

67b-3: $12x^2 + 36x$

67b-4: $6a^2 + 27a$

67b-5: $4y^2 + 36y$

67b-6: $x^2 - 6x + 9$

68a-1: $y^2 + 1y - 2$

68a-2: $5a^2 - 40a$

68a-3: $12x^2 + 36x$

68a-4: $y^2 + 2y - 8$

68a-5: $4y^2 + 36y$

68a-6: $x^2 - 6x + 9$